

# Asher Siddiqui

Karachi, Pakistan | +92-3302463301 | ashersiddique127@gmail.com | [GitHub](#) | [LinkedIn](#)

## EDUCATION

### FAST - NUCES

*Bachelor of Science in Software Engineering*

Karachi, Pakistan

*Expected Graduation: June 2025*

**Relevant Courses:** Applied DevOps, Cloud Security, Agile Software Project Management

**Awards & Honors:** Dean's List of Honors (3.5+ SGPA) in Fall 2023, Spring 2024, Fall 2024

## TECHNICAL SKILLS

**Languages:** Go, Python, Java

**Backend & Frameworks:** Echo, Spring Boot

**Databases:** MySQL, Oracle RDBMS, MongoDB, Redis

**DevOps & Cloud:** Docker, Docker Compose, Kubernetes, AWS, Linux, Bash, Prometheus, Node Exporter, Falco

## EXPERIENCE

### TAFSOL Technologies

Mar 2025 – Present

*Software Engineering Intern*

*Karachi, Pakistan (Hybrid)*

- Enabled compliance auditing by implementing logging with size based rotation for security and error logs, improving visibility and reducing storage overhead by 20%.
- Developed a QR based Quick Pay feature for internal payments, reducing average transaction time by 40%.
- Accelerated test environment setup and standardized testing across teams by deploying Docker Compose builds.
- Enhanced financial activity visibility by developing a transaction history feature, resulting in improved user trust.
- Reduced on premises hardware costs by integrating AWS S3 using AWS SDK for secure and controlled image access.

### Grayhat Private Limited

July 2024 – Sept 2024

*Software Engineering Intern*

*Islamabad, Pakistan (Remote)*

- Developed the core functionality for v0.dev's VS Code extension MVP, implementing AI generated UI component generation directly within the IDE.
- Developed an automated workflow tracker by integrating GitHub, Kimai, Activity Watch, and Plane, boosting developer productivity and visibility.

## PROJECTS

### FYP - DDoS Detection in Kubernetes using ML and DL | *Kubernetes, Ansible, Prometheus, Falco, Go, Python* | [GitHub](#)

- Deployed a bare metal Kubernetes cluster and a botnet using Ansible to create simulated DDoS attack scenarios for training ML and DL models.
- Developed a custom data collection pipeline with Falco and Node Exporter to collect system calls and resource metrics, ensuring centralized logging and data generation.
- Developed and deployed attack scripts to simulate and orchestrate attacks via a C&C server and botnet nodes, enabling automated DDoS attacks.

### Etsy Clone (Ongoing) | *Spring Boot, Docker, Docker Compose, RabbitMQ, GitHub Actions, Prometheus, Grafana*

- Built an e-commerce backend using microservices, Spring Boot and RabbitMQ, ensuring scalability of services.
- Containerized all services with Docker and Docker Compose to standardize deployment for testing and ensure consistent multiservice execution.
- Implemented CI/CD pipeline using GitHub Actions to automate Docker builds and deployments, speeding up development workflows.
- Integrated real time monitoring dashboards using Prometheus and Grafana, improving system observability and response to failures.

### Load Balancer | *Go, Express.js, Docker, Git* | [GitHub](#)

- Designed and implemented a high throughput HTTP reverse proxy implementing Round Robin and Least Connection algorithms using Go, achieving 2000 requests per second with minimal latency.
- Increased reliability by integrating health checks, ensuring automatic rerouting of traffic to available servers.
- Containerized the load balancer using Docker, enabling seamless deployment and scalability, and consistent performance across environments.